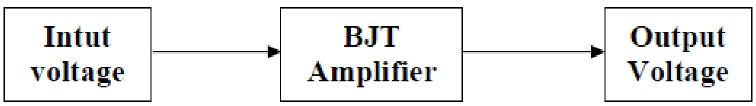
Department of Computer Science and Engineering

CSE 251 Electronic Circuits

(<u>Course Project</u>)

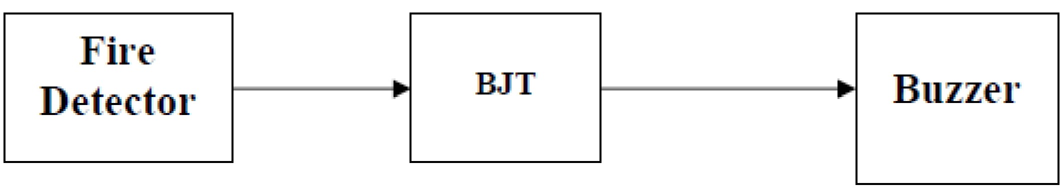
Course Outcome: CO4	Program Outcome: PO1	
Cognitive Level: C3	Psychomotor Level: P2, P3	Affective Level: A2
Knowledge Profile: K1, K3	Complex Engineering Problem: EP1, EP2	

<u>Project-01:</u> The Simplest BJT Amplifier (Odd Groups)



Design a common emitter amplifier circuit where the output voltage is increased by a minimum of two times that of the input voltage. Use the collector feedback network that employs a feedback path from collector to base to increase the stability of the system. Although the Q-point is not independent of beta (even under approximate conditions), the sensitivity to changes in beta or temperature variations is normally less than encountered for the fixed-bias or emitter-biased configurations. To build the above circuit, use the 5V peak-to-peak as input voltage and 1 kHz as frequency.

<u>Project-02:</u> A Simple Fire Alarm System (Even Groups)



Design a simple fire alarm using BJT, a diode, a fire detector, and a buzzer. For this design use a common emitter follower circuit where the input is connected with the fire detector and the output is connected with the buzzer. Show all required voltages and currents to drive the circuit properly.

Marks Distribution

Assessment Area	Mark
C3: Cognitive: Applying	4
P2: Psychomotor: Manipulation	2
P3: Psychomotor: Precision	2
A2: Affective: Responding	2
Total	**10**

The project Report should contain:

1. Problem Statement
2. Design Details
3. Circuit Diagram (Draw using **VISIO**)
4. Simulation Result